

NAVEENRAJ

Chennai, Tamil Nadu | +91 7010388606 | nraj5052@gmail.com

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PROFESSIONAL SUMMARY

Software Engineer with deep experience building and launching large-scale AI systems, backend services, and data pipelines. Experienced in setting up cloud systems, databases for multiple users, real-time AI document search (RAG), and automated infrastructure. Skilled at turning complex data into clear, natural language chat tools using smart AI orchestration and forecasting models.

CORE COMPETENCIES & SKILLS

AI & Machine Learning: AI Models (AWS Bedrock Claude/Nova, Azure OpenAI), Multi-Agent Systems, RAG Search Pipelines, Vector Search, MLflow Tracking, Time-Series Forecasting

Languages & Frameworks: Python (FastAPI, Flask, PySpark), Kotlin, Dart, JavaScript, PHP, SQL, HTML/CSS

Cloud & Data Engineering: Databricks (Delta Lake, Vector Search), AWS (Lambda, API Gateway, SQS FIFO, Secrets Manager, S3, SES), Azure (AKS, Function Apps, Service Bus, Key Vault, Cosmos DB)

Databases & Tools: PostgreSQL, MongoDB, MySQL, SQL Server, Room DB, Docker, Terraform, Git

System Design: Microservices, Multi-Tenant Databases, Platform Adapter Patterns, Async Processing, Code Parsing

PROFESSIONAL EXPERIENCE

Software Engineer – AI & Backend Systems

January 2024 – Present

Information Technology & Software Services Domain

Chennai, TN

- Build and maintain cloud-based backend systems and big data applications using Python, FastAPI, Flask, and Databricks.
- Design database structures that support multiple customer accounts with smooth, real-time switching and data updates.
- Lead the upgrade of older business processes into automated, AI-driven workflows using smart agent setups.
- Standardize backend code by creating reusable templates and adding fast, asynchronous data checks.

Junior Developer & Intern

July 2023 – January 2024

Information Technology & Software Services Domain

Chennai, TN

- Built scalable software components using clean design patterns and connected third-party communication links (APIs).
- Created data translation layers to safely connect old database systems with new front-end engines.
- Completed advanced training on software lifecycles, secure network requests, and responsive programming.

TECHNICAL PROJECTS

AI/BI Querying Engine & Predictive Forecasting Model

Current Project (2 Months)

Tech Stack: Databricks Genie, Databricks Dashboards, MLflow, PySpark, Delta Lake

- Set up a Databricks Genie space that lets users type normal questions and automatically converts them into exact, safe SQL queries across huge databases.
- Built and launched a smart sales and inventory prediction model using MLflow to track and manage different versions.
- Created live dashboards in Databricks to give stakeholders instant visual charts of real-time and predicted retail data.

Platform Orchestrator – Multi-Cloud Chatbot Deployment Automation

Project Duration: 2 Months

Tech Stack: Python, FastAPI, Terraform, AWS Lambda, SQS FIFO, Azure Functions, GCP Functions

- Built an asynchronous web receiver loop using AWS Lambda and SQS queues to separate data collection from heavy data processing tasks.
- Created an abstract adapter layer to handle data checks and format mapping across different platforms like Slack and Microsoft Teams.
- Wrote reusable Terraform code to deploy the exact same software logic smoothly across AWS, Azure, and Google Cloud (GCP).
- Programmed a smart module that automatically checks data row counts and column profiles to build custom charts instantly.

AI-Powered Service Management Platform

Project Duration: 10 Months

Tech Stack: Flask, PySpark, AWS Bedrock, Google Maps API, AWS SES, PostgreSQL, MySQL

- Built an intelligent job scheduling system using AWS Bedrock to estimate project times, calculate driving times via Google Maps, and check truck availability.
- Created a mathematical pricing tool that checks fleet use and customer demand to automatically adjust prices up or down by 5%.
- Solved multi-user database complexity by building a system that routes data safely to different database versions based on who is logged in.
- Built an automated payment reminder tool that checks overdue accounts, creates custom HTML summaries, and emails them with Excel sheets via AWS SES.

Location Resolution System – PySpark Pipeline

Project Duration: 2 Months

Tech Stack: Databricks (PySpark, Delta Lake), Databricks Vector Search, Python, Serper API

- Configured multi-threaded data workers (200 threads) to check exact name matches, falling back to a 9-table AI vector search system when confidence was below 80%.
- Engineered a backup pipeline (20 threads) that automatically scrapes company websites or searches Google via Serper API, using an AI model to clean up missing details.

Advanced Enterprise Code Intelligence Engine

Project Duration: 6 Months

Tech Stack: Python FastAPI, Azure OpenAI, AWS Bedrock, Azure Cosmos DB, MongoDB, Tree-Sitter

- Integrated multi-AI chat features using AWS Bedrock Claude and Azure OpenAI to read and explain complex code files.
- Used code parsing tools (tree-sitter) to break down raw programming files into clear, structured visual trees.
- Developed a tool to map function dependencies, tracing code paths from a single line up to the entire software system.
- Moved data smoothly from SQL Server to PostgreSQL and built fast background logging to prevent slow-downs during heavy usage.

AI-Powered Organizational Productivity Engine

Project Duration: 6 Months

Tech Stack: Python Flask, Azure Kubernetes Service (AKS), Azure OpenAI, MongoDB Vector Store

- Programmed an end-to-end document engine that reads raw files, splits them into smart sections, and stores them in a vector database for easy search.
- Created a prompt adjustment module that changes and improves AI prompts on the fly based on what the user is currently doing.
- Set up streaming answers to give users real-time, ultra-fast chat responses with zero lag.
- Moved backend services from a serverless setup into clean containers managed smoothly by Azure Kubernetes Service (AKS).

EDUCATION

Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering

August 2019 – May 2023

CK College of Engineering and Technology, Cuddalore, TN

CGPA: 8.7 / 10